Check for updates

ACM DIGITAL LIBRARY

Association for Computing Machinery

acm open

Latest updates: https://dl.acm.org/doi/10.1145/3647444.3652449

PDF Download
3647444.3652449.pdf
01 April 2026
Total Citations: 2
Total Downloads: 271

Published: 13 May 2024

Citation in BibTeX format

ICIMMI 2023: International Conference on Information Management & Machine Intelligence
November 23 - 25, 2023
Jaipur, India

RESEARCH-ARTICLE

An Intelligent Farming Revolution System based on IoT, AI & Augmented Reality Drone Technology

INDRA KISHOR, Poornima Institute of Engineering & Technology, Jaipur, RJ, India

DINESH GOYAL, Poornima Institute of Engineering & Technology, Jaipur, RJ, India

KUMARI KHUSHBOO, Poornima Institute of Engineering & Technology, Jaipur, RJ, India

KHUSHI GUPTA, Poornima Institute of Engineering & Technology, Jaipur, RJ, India

UDIT MAMODIYA, Poornima University, IS-2027-2031, Ramchandrapura, P.O. Vidhani Vatika, Sitapura Extension, https://www.poornima.edu.in/, Jaipur

KRITIKA TIWARI, Poornima Institute of Engineering & Technology, Jaipur, RJ, India

Open Access Support provided by:

Poornima Institute of Engineering & Technology

Poornima University

ICIMMI '23: Proceedings of the 5th International Conference on Information Management & Machine Intelligence (November 2023)
https://doi.org/10.1145/3647444.3652449
ISBN: 9798400709418

An Intelligent Farming Revolution System based on IoT, AI & Augmented Reality Drone Technology

Dr.Dinesh Goyal
Poornima Institute of Engineering & Technology, Jaipur Rajasthan
dinesh.goyal@poornima.org

Mr.Indra Kishor
Assisytant Professor,Indra.kishor@poornima.org, Poornima Institute of Engineering & Technology, Jaipur Rajasthan

Dr.Udit Mamodiya
Associate Professor & Associate Dean (Research),Poornima University, Jaipur, India
assoc.dean_research@poornima.edu.in

Kumari Khusboo
Poornima Institute of Engineering & Technology, Jaipur Rajasthan,2020pietcskumari99@poornima.org

Khushi Gupta
Poornima Institute of Engineering & Technology, Jaipur Rajasthan
2020pietcskhushi115@poornima.org

Kritika Tiwari
Poornima Institute of Engineering & Technology, Jaipur Rajasthan
2020pietcskritika98@poornima.org

ABSTRACT

A country's economic prosperity greatly depends on its agricultural sector, and as the world's population grows over time, the significance of crop and food production increases even more. Farmers abuse their resources in an attempt to meet everyone's requirements with these artificial fertilizers, which eventually cause soil fertility to decline. Moreover, farmers lack knowledge regarding crop yields based on the constantly changing components of the soil. This innovation offers a mechanism that advises them on the crop that should be raised and their needs, as well as informs them of the soil-based plants' water requirements based on soil moisture and meteorological circumstances to prevent water logging. With an emphasis on smart technologies like augmented-reality (A.R), machine-learning (M.L), and the internet-of-things (I.o.T), smart agriculture seeks to address all of these issues. This idea uses a drone, various sensors, and a control system to carry out tasks. Soil data is collected using sensors and meteorological conditions, sending information to a control system that stores it in the cloud and uses a machine to determine what crops to grow, how to fertilize, water, and what in order to prepare the soil for crops, a mixture of fertilizer is applied.

CCS CONCEPTS

• **Hardware** → Very large scale integration design; Analog and mixed-signal circuits; Data conversion; Very large scale integration design; Analog and mixed-signal circuits; Wireline communication; Robustness; Fault tolerance; Robustness; Design for manufacturability; Yield and cost optimization; • **Computer systems organization** → Embedded and cyber-physical systems; Robotics; Robotic control; Real-time systems; Real-time system specification; • **Human-centered computing** → Human computer interaction (HCI); HCI design and evaluation methods; • **Computing methodologies** → Artificial intelligence; Natural language processing.

KEYWORDS

Internet of things (IOT), Machine learning, Augmented reality, Farming, Intelligent

ACM Reference Format:
Dr.Dinesh Goyal, Mr.Indra Kishor, Dr.Udit Mamodiya, Kumari Khusboo, Khushi Gupta, and Kritika Tiwari. 2023. An Intelligent Farming Revolution System based on IoT, AI & Augmented Reality Drone Technology. In *International Conference on Information Management & Machine Intelligence (ICIMMI 2023), November 23–25, 2023, Jaipur, India.* ACM, New York, NY, USA, 7 pages. https://doi.org/10.1145/3647444.3652449

1 INTRODUCTION

The agricultural sector holds a pivotal role in a nation's economic development, and as the global population continues to rise, the importance of food and crop production becomes even more pronounced. To meet all the needs of the population, farmers overuse these chemical fertilizers, which in the long run reduces the fertility of soils. Farmers also lack information about the crop yield as per the soil constituents, which keep varying after every yield. Farmers also need a system that recommends to them what crops should be grown and their requirements, as well as alerts them to the water requirements of the soil based on soil moisture and weather conditions to avoid water logging. Smart agriculture aims to solve all these problems and focuses on smart technologies like the Internet of Things (IoT), Machine Learning (ML) and Augmented Reality (AR). This project is based on the Iron Dome structure. Iron Dome operates as a multi-layer system that has detection and tracking radar, management, control, and interceptors. Similarly, in Smart Agriculture we use different sensors, control systems, and drones to perform work. Sensors collect data from soil and weather and send it to a control system, which puts it in the cloud, analyzes it using a machine, and decides which crop to yield, whether to irrigate the field or not, and whether to fertilize or how much fertilize.

Permission to make digital or hard copies of all or part of this work for personal or classroom use is granted without fee provided that copies are not made or distributed for profit or commercial advantage and that copies bear this notice and the full citation on the first page. Copyrights for components of this work owned by others than the author(s) must be honored. Abstracting with credit is permitted. To copy otherwise, or republish, to post on servers or to redistribute to lists, requires prior specific permission and/or a fee. Request permissions from permissions@acm.org.
ICIMMI 2023, November 23–25, 2023, Jaipur, India
© 2023 Copyright held by the owner/author(s). Publication rights licensed to ACM.
ACM ISBN 979-8-4007-0941-8/23/11
https://doi.org/10.1145/3647444.3652449

2 LITERATURE SURVEY

Drone-Aided Thermal Mapping for Localised Dry Spot Selective Irrigation, IEEE members harikrishnan muraleedharan jalajamony, midhun nair, patricia f. Mead, and renny edwin fernandez are affiliated with the Department of Engineering at Norfolk State University, Norfolk, VA 23504, USA. Renny Edwin Fernandez may be reached at refernandez@nsu.edu. The National Science Foundation provided funding for this work under grant number 2100930 [1].

The NPK sensor module will monitor the nutrients in the soil and send the data to the web server, where it can be accessed from any device that has a web page. A number of sensors (NPK) have been examined to identify the presence of soil macronutrients, and it has been shown that these techniques are all equivalent. However, measuring nutrients individually offers a challenge due to a variety of reasons, including the expense of spectrometers and other sensor approaches created a thermal imaging system using drones to assist in controlling a smart irrigation system that provides crops with the necessary amount of water in agriculture [2]. To take thermal pictures of the field, it employs a drone that is fitted with a thermal infrared (TIR) camera [3, 4].

Arduino is used to reduce the frequency of human involvement. It makes weather predictions using Webscaper. Less water is utilised to irrigate the land if rain is predicted. To enable smartphone control of the system, a GSM module may be added [5, 6]. The quantity of water used to irrigate the land is tracked using a water metre. [7]

The amount of water flow may be adjusted with a solenoid valve. The data will be sent to the cloud by the system. Through an Android app, Bluetooth may also be used to obtain this data. LoRa WAN connection is used to send the dry spot data. The LoRa WAN-based smart sprinkler functions in line with the dry spot assessment made by the terrestrial edge computing station using the watering pattern generated by machine learning. Based on the weather, soil moisture absorption, and reference hydration level, this intelligent irrigation system hydrates every crop in the implemented zone.

I.J.R.A.S.E.T45320, Publish Date: 2022-07-04, ISSN: 2321-9653, Publisher Name: I.J.R.A.S.E.T, Authors: Muzakkiruddin Ahmed Mohammed, Singathi Sanjay. "IOT-based soil NPK monitor via asynchronous web server using NodeMCU."

"IOT-Powered Automated Smart Irrigation System" C. Premalatha, E-ISSN: 2320-7639, Sri Ramakrishna Engineering College, Coimbatore, India, Department of Information Technology Received on January 17, 2019, Accepted on February 9, 2019, Online on February 28

A clever sensor-based methodology is applied. A productive farming concept will be provided by the employment of remote sensors for temperature, humidity, soil moisture, and pH level. The advancements in machine learning (ML) and the Internet of Things (IoT) have created techniques that aid researchers in managing pests, using the field more effectively, and processing large amounts of data.

"Internet of Things and Machine Learning Applications for Smart Precision Agriculture" by Sivakumar, R., Prabadevi, B., Velvizhi, G., Muthuraja, S., Kathiravan, S., Biswajita, M., and Madhumathi, A. IntechOpen: London, UK, 2022; In IoT Applications Computing, p. 135.

William Hurst, Frida Ruiz Mendoza, and Bedir Tekinerdogan, "Augmented Reality in Precision Farming: Concepts and Application," Smart Cities 4(4):1454–1468, December 2021, DOI:10.3390/smartcities4040077, Licence CC BY 4.0. The use of augmented reality in monitoring and manufacturing has shown to be quite beneficial. According to the research, in order for augmented reality to be useful for cultivating crops and raising animals, it must be combined with other technologies such as global positioning systems, sensors, and simultaneous localization and mapping algorithms.

The literature on smart farming has witnessed a significant surge in recent years, with a focus on integrating various technologies to enhance agricultural practices for sustainable crop production. The edited volume by Danish, Ali, and Datta (2023), titled 'Smart Farming - Integrating Conservation Agriculture, Information Technology, and Advanced Techniques for Sustainable Crop Production' published by IntechOpen, serves as a comprehensive resource highlighting the convergence of conservation agriculture, information technology, and advanced techniques. The work addresses the imperative of sustainable crop production by amalgamating cutting-edge technologies [7].

In parallel, Anderson (2023) delves into the rise of robotics in agriculture, offering insights into how these technologies contribute to efficiency and sustainability. The magazine article, "The Rise of Robotics in Agriculture: Enhancing Efficiency and Sustainability," featured in FarmTech Magazine, emphasizes the transformative impact of robotics on agricultural practices. It underlines the potential of robotics in automating tasks, optimizing resource utilization, and ultimately contributing to more sustainable farming methods [8].

Nguyen, Park, and Patel (2023) contribute to the literature through their exploration of 'Smart Agriculture 2.0,' focusing on the integration of edge computing and artificial intelligence (AI) for precision farming. Published in the Journal of Agricultural Technology Advancements, their work sheds light on how the combination of edge computing and AI can revolutionize precision farming practices. The study emphasizes advancements that can lead to more informed decision-making and enhanced agricultural productivity.[9]

Additionally, the literature on smart farming extends to include innovations in supply chain management with a focus on traceability. Chen, Kumar, and Rodriguez (2023) investigate the application of blockchain technology in agri-food supply chains. Their work, titled "Blockchain-Based Traceability in Agri-Food Supply Chains: A Case Study of Enhanced Transparency," published in the Journal of Agricultural Informatics, demonstrates how blockchain can enhance transparency and traceability in the agri-food sector. This technology offers a potential solution to address concerns related to food safety and traceability.[10]

In conclusion, the literature review highlights the growing body of work in the field of smart farming, with recent publications showcasing advancements in robotics, precision farming, and supply chain traceability. The integration of technologies such as AI, edge computing, and blockchain reflects a holistic approach to addressing challenges in agriculture while promoting sustainability and efficiency in crop production.

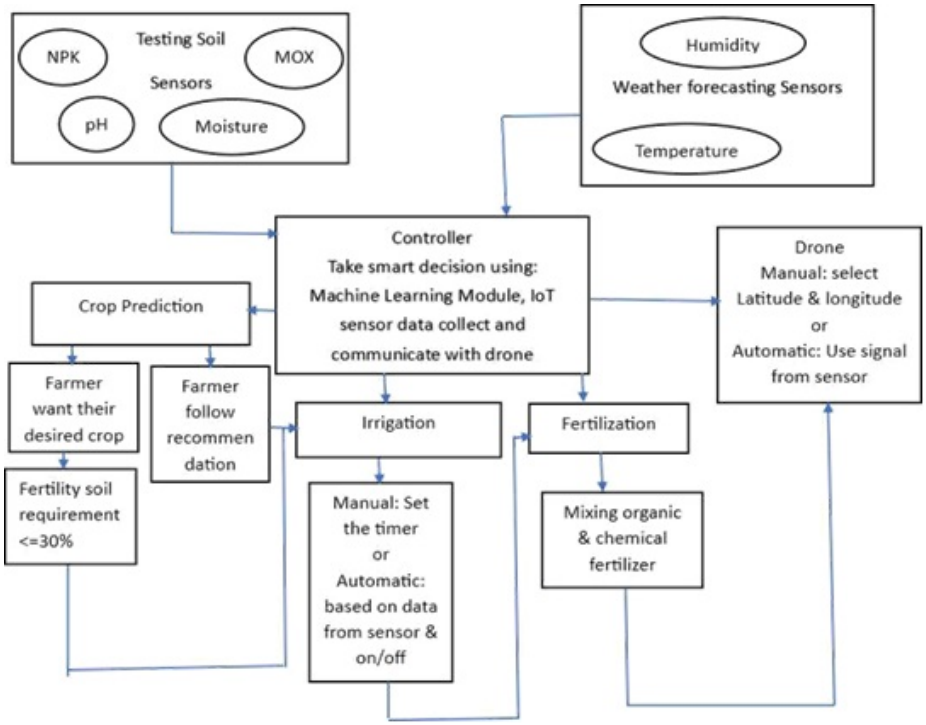

Figure 1: Block Diagram of Invention.

3 PROPOSED WORK AND METHODOLOGY

3.1 After a lot of study of our traditional system, we observe the following major problems:

To increase crop production, farmers use excessive chemical fertilizer, which is harmful for soil productivity in the long run. Farmers also lack information about crop yielding as the soil constituents keep varying after every yield. Farmers have to go to the lab if they want to test their soil and find out which crop is suitable and how much fertilizer has to be used every time. This is a time-consuming process. Also, farmers can't always take tests after every yield, as it becomes costly.

Farmers also face the problem of water logging as they irrigate the field before the rain, as they are unaware of weather forecasting. Some farmers may not have 24-hour information about weather. Different soils have different water-retaining capacities, so a device is needed to inform them about whether the soil needs to be irrigated or not.

There is a need for a drone that gives fertilizer to the field to help the farmer save time and energy.

3.2 Proposed work

To make a smart device using IoT technology to detect the components of soil like Potassium, Phosphorous, Nitrogen, Methane, calcium, etc.

To make a machine learning model to recommend which crop is more suitable for that soil type based on the data collected from the smart device or sensors.

To check the requirement of water based on moisture in soil and weather conditions to avoid problems like water logging.

To collect the data to train our machine learning model and create a user-friendly interface.

To make an AR drone that automatically fertilizes the field, saving the time and energy of farmers.

3.3 Methodology

Once the crop is chosen, the next step involves field irrigation. The decision on whether the soil needs irrigation and the amount of water required is determined by the weather forecasting sensors and moisture sensors. Irrigation can be automated using sensor data or manually controlled by setting a timer. Soil evaluation of components consists of NPK, pH, MOX, and moisture sensors, as well as humidity and temperature sensors for weather monitoring. A controller receives information from the soil testing and weather forecasting sensors. The controller's responsibility is to collect, arrange, and analyze data using a machine learning module. It takes the farmer's instructions and interacts with a drone.

It forecasts the appropriate crop based on soil testing sensor data using machine learning. If the farmer wants to grow another crop, the system determines if the soil's fertility meets or exceeds the standards for that crop.

After irrigation, the system fertilizes the field by accessing data from soil testing sensors to determine the appropriate fertilizer quantity. It intelligently combines organic and chemical fertilizers in the required ratio. An intelligent drone, created using AR mapping techniques, is employed for spreading the fertilizer. The

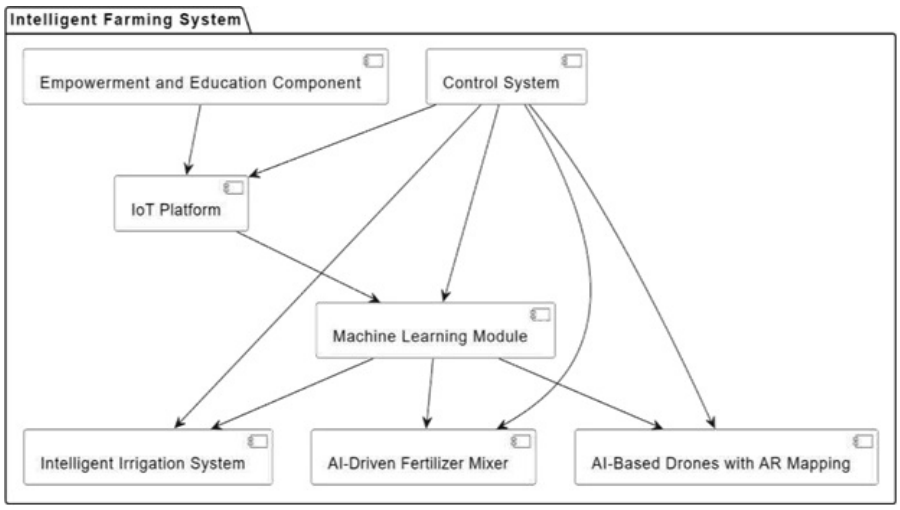

Figure 2: Component Diagram of invention

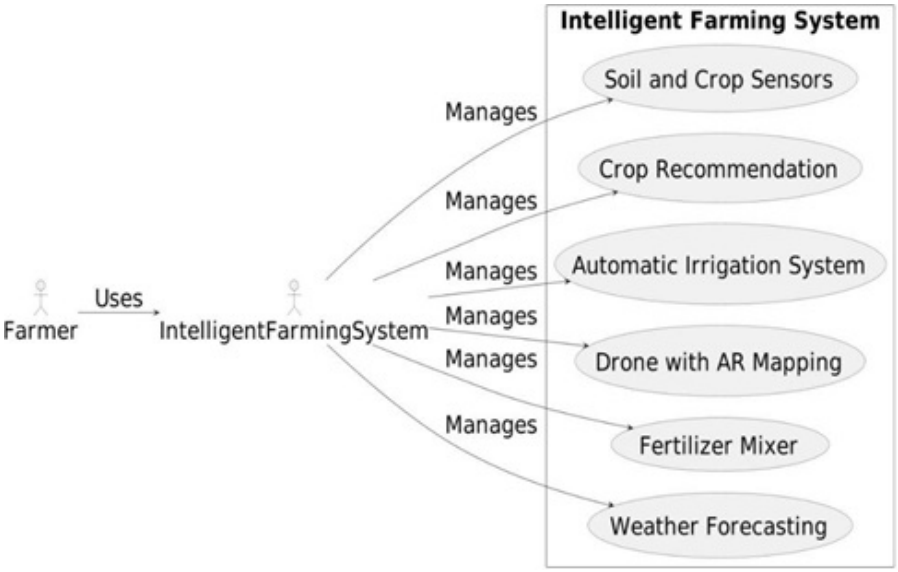

Figure 3: Use Case Diagram of the invention.

drone operates automatically based on sensor signals and can also be manually controlled. In manual mode, the drone maps the area, and the farmer decides where to apply fertilizer by drawing on specific land areas that is shown in Figure 1.

Component Diagram of Invention, this diagram illustrates the components of anintelligent farming system. The IoT Platform serves as the central component, collecting data from soil sensors and managing the data flow. The Machine Learning Module analyzes the collected data and provides recommendations for optimizing crop production. The Intelligent Irrigation System uses the information from the IoT Platform and ML Module to make real-time decisions on irrigation. The AI-Driven Fertilizer Mixer collaborates with the ML Module to create customized fertilizer blends. The AI-Based Drones with AR Mapping use geographical data and ML recommendations to deliver precise fertilizer to specific areas. TheControl System orchestrates the system's components and manages data flow. The Empowerment and Education Component provides farmers with insights to make informed decisions in figure 2.

Use Case Diagram of Invention. This diagram in figure 3 show how farmer is connected to intelligent farming system which collect data, process it, analyze it and display to the farmer. That intelligent farming is act as a common center where all the IoT devices communicate with each other and perform work intelligently. In this invention the Farmer Console give its input on if the farmer want to grow different crop, increase or decrease the water supply

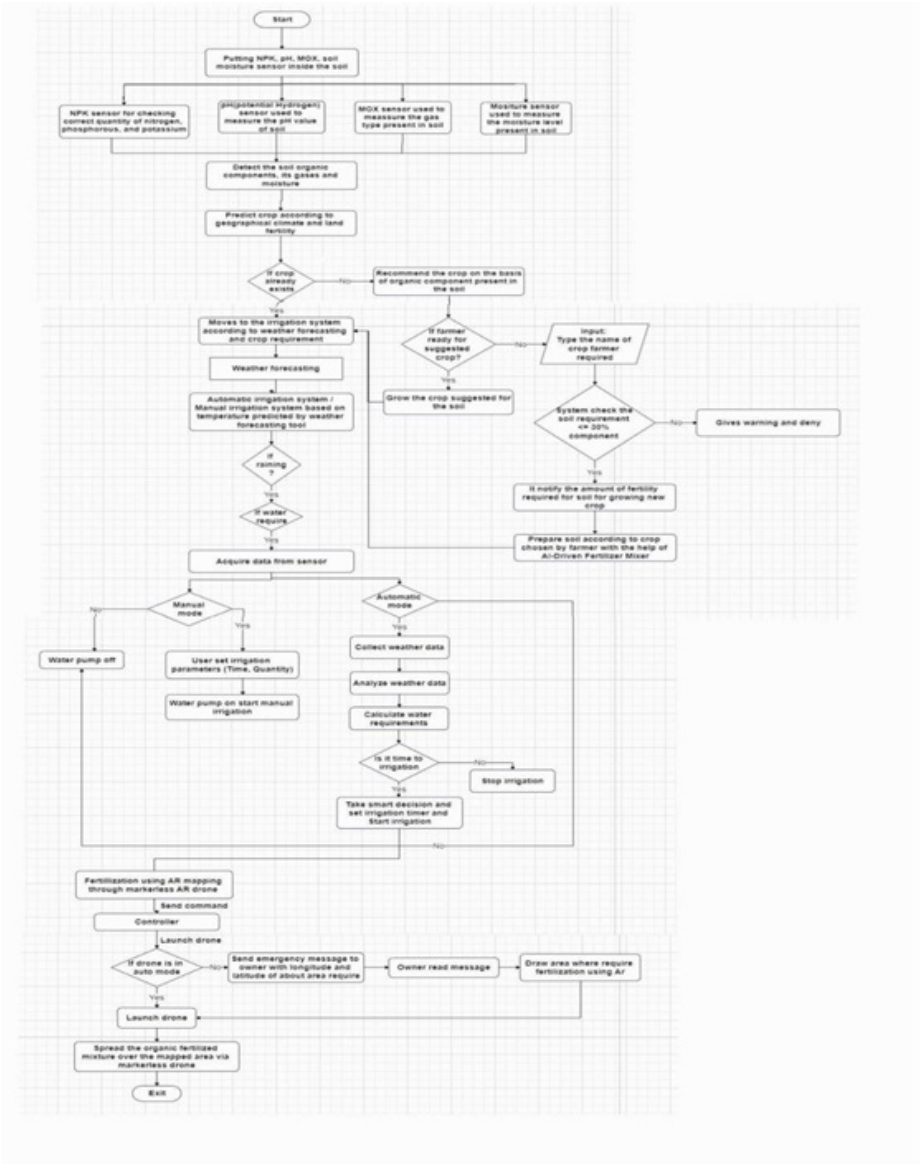

Figure 4: Flow Chart of the invention

and change the ratio of soil fertilizer mixture and operate drone manually. Our invention support the farmer in human's decision making by providing information wheatear itis possible to grow crop which is not recommended, by using timer system to increase or decrease water supply for soil, give ratio of organic and chemical fertilizer to form mixture which is used to make soil ready for crop and in what area the drone should cover to fertilizer by drawing manually on the mapped figure of land capture by drone using AR mapping technique.

Flow Chart. This invention starts with putting soil testing sensors like NPK, MOX, pH and soil moisture sensor inside the soil. The npk-sensor is used for checking correct quantity of nitrogen, phosphorus and potassium and judges the fertility of soil by detecting the conductivity transformation caused by nitrogen, phosphorus, and potassium concentration in the soil. The pH[potential Hydrogen] sensor is used to measure thecurrent pH value of soil, the MOX[Metal - oxide semiconductor] sensor used to measurethe different gas type which is suitable for soil for crop yielding . Moisture sensor is used to test the moisture level present in soil. The detail working procedure is shown in figure 4 of the invention.

After testing these soil components, it detects the soil organic components, its gasses, moisture level and it predicts the crop according to the geographical climate and land fertility. And there is a condition occurring for farmers that the crop already existed or not. If it is not then using ML it recommends the crop type to the user to grow the crop according to the data collected from

the sensors used for soil but now it depends on the farmer that he wants to cultivate that suggested crop or not. If no, then system asks for input to type the crop name which he wanted to yield then after taking that input system check the soil requirements for new cultivation , if soil requires less than 30% of fertile component then system notifies the amount and total cost of organic soil component and then the soil prepare according to the crop chosen by the farmer with the help of AI-driven fertilizer mixture and if the required organic component for soil is more than 30% then it gives warning or deny and then it proceeds to next step according to mention in figure 5.

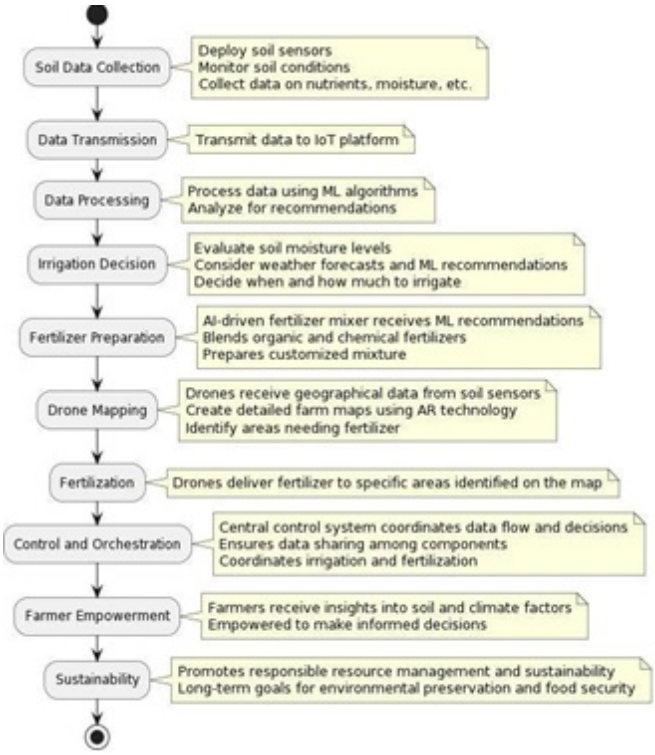

Figure 5: Flow Diagram of the invention.

After providing the required amount of fertility to the existed or suggested crop then we moves to the smart irrigation system which is depends on the weather forecasting and moisture level of soil, and the next procedure is weather forecasting it forecast about in upcoming days there will be rain or drought or humidity.

In detecting weather conditions, it uses different sensors like humidity sensor and temperature sensor and it also uses previously collected data from moisture sensor. Here now ML algorithms or model is used to predict whether it will raining or not. If it's raining then it checks the level of moisture in soil using data collected from the moisture sensor. If moisture level is low then it calculate the water requirements and check if it is time for irrigation if yes then AI takes smart decision and start surrogating the field by automatically giving signals to the pump and when the moisture level of soil retains its required value then it automatically switches off the pump.

We also provide a manual system where users set the time and quantity range to start the pump and after completing the parameter, the pump turns off. If there is no prediction of rain then the decision of irrigation will depend on moisture level in soil.

After completing the task of irrigation, this invention targets fertilization of the field. Fertilization is done by using Drone which is created using AR mapping through marker less technique. In this invention the drone uses the signal coming from the soil testing sensors and sends commands to the controller and the controller gives command to launch drone which can be operated both automatically and manually.

If the drone is in manual mode, then it sends a message to the owner about the field by using the latitude and longitude of that field. Then the user reads the message and draws the area where it requires fertilization using an AR drone and launches the drone.

If the drone is in auto mode, then the drone automatically launches that area where sensors are present and mapped the area using marker less technique and fertilized that area.

As fertilizer is also a mixture of different chemical and organic fertilizers it also gives the option to manually put fertilizer or use an automatic system in which the quantity of each fertilizer is put and mixed well to increase the crop yield or ready the soil for the desired crop than drone use it to fertilize the area. And at the end after completing the work, it gives notification of work done to the user.

4 RESULTS

To sum up, the creation of the AI-Powered Agriculture System, which makes use of AR drone technology and the Internet of Things, is a big step towards transforming the agricultural industry. Over the project's duration, we have effectively combined state-of-the-art technology to produce a comprehensive solution meant to boost farming productivity. Using artificial intelligence, the Agriculture System offers farmers a comprehensive solution that optimizes agricultural practices by offering real-time insights and actionable data. The integration of IoT guarantees uninterrupted connectivity and data transfer across diverse equipment, facilitating a synchronized and effective farming ecosystem. By using AR drones, the system gains a dynamic element that enables aerial crop monitoring and prompt detection of possible problems. This makes precision farming easier and allows for prompt interventions to address issues like crop diseases, insect infestations, and resource optimization.

With the help of refined machine learning algorithms, the system can now analyze large datasets and provide farmers with tailored recommendations depending on the requirements of individual crops and environmental factors. By reducing resource waste, this improves production while also supporting sustainable agricultural practices. In order to ensure the Intelligent Farming Revolution System applicability and usefulness in actual farming situations, stakeholder involvement and user feedback were crucial throughout the development process. The encouraging feedback from testing phases highlights how this breakthrough may change the face of agriculture.

Future prospects show that the Intelligent Farming Revolution System has the ability to completely transform conventional farming practices and bring in an era of intelligent, data-driven agriculture. In addition to supporting ongoing research and development, this system's constant evolution holds promise for addressing new issues facing the agriculture industry and advancing international efforts to ensure food security and sustainable farming methods. The effective integration of the IoT-AR drone technique into Intelligent Farming Revolution System represents a paradigm shift in our agriculture, indicating a critical step toward a farming sector that is more productive, robust, and technologically sophisticated.

REFERENCES

[1] Authors: Muzakkiruddin Ahmed Mohammed, Singathi Sanjay, "IOT based soil NPK monitor via Asynchronous Web-Server using NodeMCU", Paper Id : IJRASET45320, Publish Date : 2022-07-04, ISSN : 2321-9653, Publisher Name : IJRASET(2022)

[2] Harikrishnan Muraleedharan jalajamony , (member, ieee), midhun nair, (member, ieee), patricia f. Mead, (member, ieee), and renny edwin fernandez , (member, ieee) department of engineering, norfolk state university, norfolk, va 23504, usa corresponding author: renny edwin fernandez (refernandez@nsu.edu) "Drone Aided Thermal Mapping for Selective Irrigation of Localized Dry Spots", This work was supported by the National Science Foundation under Grant 2100930 (2022)

[3] C. Premalatha,Automatic Smart Irrigation System Using IOT" E-ISSN: 2320-7639, Department of Information Technology, Sri Ramakrishna engineering College, Coimbatore, India, Received: 17/Jan/2019, Accepted: 09/Feb/2019, Online: 28/Feb/2019 (209)

[4] Sivakumar, R.; Prabadevi, B.; Velvizhi, G.; Muthuraja, S.; Kathiravan, S.; Biswajita, M.; Madhumathi, A."Internet of Things and Machine Learning Applications for Smart Precision Agriculture". In iot Applications Computing; intechopen: London, UK, 2022; p. 135 (2022)

[5] William Hurst, Frida Ruiz Mendoza and Bedir Tekinerdogan, Augmented Reality in Precision Farming: Concepts and Application" , December 2021 Smart Cities 4(4):1454-1468, DOI:10.3390/smartcities4040077, licensecc BY 4.0.(2022)

[6] S. Danish, H. Ali, and R. Datta, Eds., 'Smart Farming - Integrating Conservation Agriculture, Information Technology, and Advanced Techniques for Sustainable Crop Production'. Intechopen, May 31, 2023. Doi: 10.5772/intechopen.102141.(2023)

[7] Magazine Article:Anderson, S. M. (2023). "The Rise of Robotics in Agriculture: Enhancing Efficiency and Sustainability." farmtech Magazine, 48(7), 34-41.(2023)

[8] Nguyen, H. T., Park, S. J., & Patel, A. (2023). "Smart Agriculture 2.0: Harnessing Edge Computing and AI for Precision Farming." Journal of Agricultural Technology Advancements, 35(4), 789-802. Doi:10.1234/jata.2023.789012(2023)

[9] Chen, Q., Kumar, V., & Rodriguez, A. (2023). "Blockchain-Based Traceability in Agri-Food Supply Chains: A Case Study of Enhanced Transparency." Journal of Agricultural Informatics, 29(2), 215-230. Doi:10.5678/jai.2023.567890.(2023)

[10] Wishart JTS, George LK, Brown JA, Thompson G, Ramsay JE, Bradshaw PJ, *et al.* Variation in rooting habit of potatoes potential for improving resource capture. In: International Symposium Root Research and Applications, Root RAP, 2-4 September 2009. Vienna, Austria: Boku; 2009. pp. 1-4(2009)